

2-Mark Questions & Answers -- Data Structures Module 4: Trees

Q1. Define a Binary Tree and list its representations.

A Binary Tree is a hierarchical non-linear data structure where every node has at most two children (left and right).

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Q2. What is a Full Binary Tree vs Complete Binary Tree

- * Full Binary Tree: Every node has either 0 or 2 children.

* Co

Q3. Define Balance Factor in an AVL Tree.

The Balance Factor (BF) of a node in an AVL tree is defined as:

Balance
Valid

Q4. What is an Expression Tree

An Expression Tree is a binary tree representation of an arithmetic expression where internal nodes are operators (+, -, *, /) and leaf nodes are operands (constants or variables).

Q5. Differentiate between B-Tree and B+ Tree.

- * B-Tree: Stores keys and data record pointers in both internal and leaf nodes.

* B+